GLOBALTECH

Cybersecurity Internship · Cohort 1

Week 2 Mission Brief

OPERATION HOLLOW BANK

Network Traffic Analysis & Incident Response

Programme	GlobalTech Cybersecurity Internship — Cohort 1
Client (simulated)	TC.Solutions Fintech
Operation codename	HOLLOW BANK
Week	Week 2 of 8 (Milestone 1, Month 1)
Window	Monday 13 July – Friday 17 July 2026
Submission deadline	Friday 17 July 2026, 18:00 WAT
Results published	Sunday 19 July 2026, 18:00 WAT
Evidence file	Wireshark_PCAP.pcap (provided — open before you read anything else)

Setting

TC.Solutions Fintech is a Nigerian payment processing company and a GlobalTech client. They handle card transactions, mobile transfers, and API-based payments for over 40,000 active customers across Nigeria. They are regulated by the Central Bank of Nigeria (CBN) and are subject to PCI DSS v4.0, the NDPA 2023, and the CBN Risk-Based Cybersecurity Framework — frameworks you mapped in your D1 research last week.

On Monday morning, before boarding a flight for a two-week leave, GlobalTech SOC Manager Nsikan Ubong logged into the TC.Solutions SIEM and flagged an anomaly on one internal host. He wrote a single handover note:

> *Something is off with 192.168.1.105. Unusual outbound connections — destination I don't recognise. Pulled a PCAP but haven't had time to go through it. Attached. Cover this until I'm back. — Nsikan*
>
> *Download the PCAP here.*

You are the Tier-1 analyst on shift. By Tuesday morning, three more SIEM alerts have fired on the same host. You open the packet capture.

What you find is worse than Nsikan thought.

Your Environment

Asset	IP Address	Role
Internal Host A	192.168.1.100	Staff workstation — finance department
Internal Host B	192.168.1.105	THE FLAGGED HOST — source of anomalous traffic
Internal Host C	192.168.1.110	Staff workstation — IT/API team
Internal Host D	192.168.1.120	Staff workstation — operations
Internal Server	192.168.1.200	Internal API server — fintechcorp.ng
Gateway	192.168.1.1	Network gateway
External — SUSPICIOUS	45.33.32.156	Unknown external IP — flagged in Nsikan's note
External — DNS	8.8.8.8	Google Public DNS
External — Nigeria	197.255.36.10	Nigerian external IP
External — Africa	196.201.214.5	African external IP
External — Google	172.217.20.141	Google (accounts.google.com)

⚠ EVIDENCE REQUIREMENT

Open the PCAP file before you write a single sentence. Every claim in D1, D2, and D3 must cite a specific packet number (e.g. Pkt 61), protocol, IP address, port, or payload extract from the capture. Assertions not traceable to the PCAP will be marked as unsupported and will fail.

What the PCAP Contains — Analyst Orientation

You are not told what to find. You are told where to look. The PCAP contains 122 packets spanning multiple protocols and hosts. Below is a structural orientation — not the answer. Your job is to read the packets, identify what is suspicious, build the evidence table in D1, and explain what happened.

Packet Structure Overview

Packet Range	Protocol(s)	What to Investigate
Pkts 0–15	UDP / DNS	Eight DNS queries from four internal hosts. Who queried what? Are all domains legitimate?
Pkts 16–57	TCP / TLS (443)	Encrypted HTTPS sessions from multiple hosts to external IPs. Note which IPs and what the session patterns look like.
Pkts 58–77	TCP / HTTP (80)	Unencrypted HTTP traffic. Read the payloads carefully. What host? What destination? What data was sent?
Pkts 78–93	TCP (port 4444)	Eight sequential outbound connections from one host to the suspicious external IP. Read the payload bytes. What does this look like?
Pkts 94–110	TCP / HTTPS / HTTP	Additional external sessions and an internal API call. Note the Bearer token.
Pkts 111–116	ICMP	Three ping sequences from one internal host to three different targets. What is this host doing?
Pkts 117–122	TCP	Connection teardowns. Note which sessions close cleanly.

INVESTIGATOR NOTE

Port 4444 is the default listener port for Metasploit Framework's Meterpreter reverse shell payload. Its presence in outbound traffic from an internal host — especially in sequential, patterned connections — is a significant indicator of compromise. Research this before writing D1.

D1 — Network Traffic Analysis Report

Length: 5–7 pages

Addressed to: Internal — for SOC record and handover to Nsikan Ubong on his return

Audience: Technical — your fellow analysts and your SOC manager

Purpose

Prove there is a real security incident by pinning each suspicious event to specific packet evidence with a confidence assessment. This is the foundational document — D2 and D3 are built on what you find here.

Required Structure

Page 1 — Methodology

- State your analysis approach: how you worked through the PCAP, what tools you used or would use (Wireshark, tshark, scapy, or equivalent), and what you were looking for.
- Define your confidence scale — High / Medium / Low — with one example of each from the capture.
- State the scope: which hosts, which time window, which protocols you examined.

Pages 2–5 — Evidence Table

Build an evidence table with a minimum of 10 rows. Each row must cite at least one specific packet by number. A short analyst commentary paragraph (3–5 sentences) must appear directly below each row explaining the significance of that event in the context of the broader incident.

Table columns:

Column	What to put in it
#	Row number
Packet(s)	Specific packet number(s) from the PCAP — e.g. Pkt 61, Pkts 78–93
Timestamp / Protocol	Protocol and layer (e.g. TCP/HTTP, UDP/DNS)
Source → Destination	IP:port → IP:port
Observation	What you observed — specific, factual, no conclusions yet
Corroborating evidence	At least one other packet or external reference that supports this observation
Confidence	H / M / L with one-line justification

Mandatory rows — these must appear in your table:

- **Row 1:** The DNS query to the lookalike domain (Pkt 2 or Pkt 4) — cite the queried name verbatim.
- **Row 2:** The HTTP GET to the suspicious host (Pkt 58) — state the full URL and destination.
- **Row 3:** The credential submission (Pkt 61) — cite the HTTP method, destination, and what data was transmitted. Do NOT reproduce the password in full — refer to it as 'cleartext credentials observed in payload'.
- **Row 4:** The first C2 beacon (Pkt 78) — cite the destination IP, port, and the pattern string visible in the payload.
- **Row 5:** The ICMP sweep (Pkts 111–116) — identify which host performed it and what three targets were probed.

Rows 6–10 are yours. Defensible choices: the sequential nature of the 8 beacon connections, the internal API call with Bearer token, the fake session cookie, the HTTP response content pattern, the DNS query to internal.fintechcorp.ng.

Page 6 — Pattern Summary

- Name the attack pattern you observed — what kill chain stage does this represent?
- Identify the single most critical row in your table and explain why a grader should reread it first.
- State your next investigative step: if you had 30 more minutes before shift handover, what would you do?

D1 Rubric

Criterion	What graders check
No generic answers	Every row cites a specific packet number. 'Unusual traffic was observed' without a packet reference = zero for that row.
Mandatory rows present	All 5 mandatory rows present with correct packet citations. Missing a mandatory row = automatic 5-mark deduction.
Corroboration	Every row has at least one corroborating packet or external reference (MITRE technique ID, CVE, or RFC).
Commentary quality	Analyst commentary shows understanding of WHY the event is suspicious — not just WHAT happened.
Confidence assessment	H/M/L scale defined on Page 1 and applied consistently. A row marked 'High' must have stronger evidence than one marked 'Low'.

D2 — Incident Report to the CISO

Length: 3–5 pages

Addressed to: Prosper Eromonsele, CISO, GlobalTech

Audience: Non-technical executive — write for someone who will make a decision based on your report in the next 15 minutes

Purpose

Translate your D1 technical findings into a formal incident report that the CISO can act on. No MITRE technique IDs in prose — those go in D3. Plain language. Every claim still cites evidence — but evidence goes in an appendix, not the main text.

Required Structure

1. Headline Finding — first sentence states the conclusion. Example structure: 'TC.Solutions Fintech was compromised on [date] by [attack type] originating from [source]. The affected host is [IP]. The attacker established [what].' Write this paragraph last.
2. Situation Summary — one paragraph the CISO could read aloud at a board meeting. No jargon. Names the client, the affected systems, and the nature of the threat without technical detail.
3. What Is at Risk — what data classes were touched or potentially exfiltrated? What does GlobalTech know about the attacker? What cannot yet be ruled out? Anchor every claim to the PCAP.
4. Three 72-Hour Actions — one paragraph each (60–100 words). Each action must be: verb-first, owner role (not a name), hour-precise deadline, evidence cite (≥2 packets or D1 rows), and the ONE risk the action does NOT close.
5. Systemic Recommendation + Evidence Appendix — one policy or process recommendation (not a tool ask). Appendix: bulleted list of every packet, IP, domain, and external reference cited across D2.

CITATION BAR

Every action in Section 4 must cite at least 2 packets or D1 rows. The systemic recommendation must cite at least one framework from your Week 1 D1 research. The appendix must be exhaustive — if you cited it anywhere in D2, it appears here.

D2 Rubric

Criterion	What graders check
Headline finding	First sentence of Section 1 states a clear, evidence-backed conclusion. Vague opening ('We detected suspicious activity') = fail for this criterion.
Plain language	No MITRE IDs, no technical jargon in the main text. A non-technical executive can read every sentence without a glossary.
72-hour actions	All three actions are verb-first, have an owner role, a deadline, ≥2 evidence cites, and name one risk NOT closed.
Framework reference	Systemic recommendation cites at least one framework from Week 1 D1 by name (e.g. NDPA 2023 breach notification, CBN RBCSF incident classification).
Appendix completeness	Every IP, domain, packet, and external reference cited in D2 appears in the appendix. Missing entries = deduction per missing item.

D3 — MITRE ATT&CK Mapping

Length: 2–3 pages

Format: Mapping table + narrative summary

Purpose

Map every observed behaviour in the PCAP to a named MITRE ATT&CK technique. This is the threat intelligence layer of your investigation — it tells GlobalTech and TC.Solutions what class of attacker they are dealing with and what that attacker's next move might be.

Required Structure

Part A — ATT&CK Mapping Table

Minimum 6 rows — one per observed technique. Each row must cite the packet(s) that evidence the technique.

#	Tactic	Technique Name	Technique ID	Evidence from PCAP (packet #)	Analyst Note
Ex.	[fill in]	[fill in]	T1XXX	Pkt [X]	[fill in]
2					
...					
6+					

Mandatory techniques — these must appear in your table:

- T1566 — Phishing (or sub-technique): the lookalike domain and credential harvesting page
- T1071 — Application Layer Protocol: the C2 communication over TCP port 4444
- T1078 — Valid Accounts: the use of submitted credentials to access the attacker's portal
- T1046 — Network Service Discovery: the ICMP sweep across three targets

Rows 5 and 6+ are yours. Defensible additions: T1041 (Exfiltration over C2), T1059 (Command and Scripting Interpreter), T1190 (Exploit Public-Facing Application), T1040 (Network Sniffing).

Part B — Narrative Summary (1 page)

Write a one-page narrative that walks through the attack chain in chronological order, referencing your ATT&CK table by technique ID. Explain what the attacker did, in what order, and what their likely objective was. Name the kill chain stage at each step (Reconnaissance, Weaponisation, Delivery, Exploitation, Installation, Command & Control, Actions on Objectives).

D3 Rubric

Criterion	What graders check
Mandatory techniques present	All 4 mandatory techniques correctly identified, named, and cited to specific packets.
Technique ID accuracy	IDs match the current MITRE ATT&CK framework (v15). Wrong or invented IDs = fail for that row.
Evidence citation	Every row cites at least one specific packet number. Technique without packet evidence = unsupported assertion.

Criterion	What graders check
Narrative coherence	The Part B narrative tells a chronological story connecting the techniques into a complete attack chain. Bullet-point lists are not a narrative.
Kill chain alignment	Each narrative stage is correctly labelled with the corresponding kill chain phase.

D4 — Judgment, Briefing & Public Communication

D4 has three parts: a written judgment essay, a video briefing for your supervisor, and a video post for your professional community. All three are mandatory. Missing any one part means D4 is treated as incomplete.

Part A — Judgment Essay (1 page, 300–400 words)

You are reviewing Pkt 61. In the payload, cleartext credentials are visible: a username and a password submitted via HTTP POST to a suspicious external domain. You now know — from your D1 analysis — that this host was subsequently beaconing to the same external IP on port 4444.

Your direct manager is Nsikan Ubong. He is on leave and unreachable by phone. The CISO, Prosper Eromonsele, is in back-to-back client meetings until 17:00. It is 14:30. You have the credentials in front of you.

> *A colleague on the SOC floor leans over and says: 'Those are live credentials for a GTBank lookalike site. We could log in and see what the attacker has collected. It would give us better threat intelligence.' What do you do?*

Address all four:

6. What you would do — step by step, in the next 2 hours. Be specific about who you contact, what you document, and in what order.
7. What you would NOT do, and why — specifically address your colleague's suggestion. Name the legal or ethical principle that applies.
8. The ISC2 canon (by name) that governs your decision. Explain how it applies to this specific situation, not generically.
9. What you write in the shift handover note for the next analyst — the exact content, not a description of it.

HONESTY RULE

Two graders read every D4 Part A independently. If both flag the writing as untraceable to your own PCAP analysis — no packet references, no specific details from this capture — Part A is treated as not submitted.

Part B — Video 1: Supervisor Briefing (minimum 2 minutes)

Record a professional video briefing addressed directly to Nsikan Ubong. He is on leave and cannot attend a call — this video is how you bring him up to speed on what you found in his PCAP.

What your video must cover:

- Who you are, your role, and the context (Nsikan flagged this before he left)
- What you found — the key events from the PCAP in chronological order, referenced by IP, port, or packet where relevant
- Your confidence level in the findings and what remains uncertain
- What actions you have already taken or recommended in D2
- What you need from Nsikan when he returns or is able to respond

Professional standards for this video:

- Address Nsikan directly — 'Nsikan, I'm reaching out because...' — not a generic presentation
- Minimum 2 minutes. No maximum — but don't pad. Every sentence should be there for a reason.
- Professional appearance and setting. This is a work briefing, not a casual vlog.

- No reading from a script verbatim — you may use notes, but the delivery must be natural and confident
- **Confidentiality:** do not name TC.Solutions Fintech's real client data, real account numbers, or reproduce credentials in your video. Reference 'the affected host' or 'the suspicious external IP' — not specific internal IP addresses if you are recording in a public setting.

Part C — Video 2: Community & LinkedIn Post (minimum 2 minutes)

Record a second video — different in tone, same in quality — for your professional community. This video accompanies your LinkedIn post this week. It should speak to peers, students, and the cybersecurity community, not to your supervisor.

What your video must cover:

- What this week's task involved — without disclosing client details or specific attack findings
- What you learned — specific skills applied (network traffic analysis, MITRE ATT&CK mapping, incident reporting)
- How this work connects to the frameworks you researched in Week 1 (name at least one)
- What value you can now offer — what can you do today that you could not do confidently two weeks ago?
- A genuine reflection on GlobalTech's role in building this capability in you

LinkedIn post requirement:

Alongside Video 2, submit a LinkedIn/X post of 150–250 words. The post must: embed or link Video 2, describe what you worked on this week (without client specifics), tag your lecturer, the Director, and @GlobalTechWorldComputerInstitute, and use relevant hashtags (#Cybersecurity #NetworkForensics #MITRE #GlobalTech #Internship).

CONFIDENTIALITY GUARDRAIL — READ BEFORE RECORDING

Video 2 is public. Standard industry practice: never name real or simulated client details, specific IP addresses, credentials, or attack findings in a public video. Describe your work in terms of skills and methodology — not case specifics. This is the habit that protects you and your clients throughout your career.

D4 Rubric

Criterion	What graders check
Part A — Judgment reasoning	Clear position taken; trade-off acknowledged; specific to this PCAP (packet references present); not a policy recitation.
Part A — ISC2 canon named	Correct canon named by title and applied specifically to the colleague's suggestion — not generically.
Part A — Handover note	Actual note content written out — not described. Specific, actionable, references the incident.
Part B — Video 1 quality	Addresses Nsikan directly; covers all 5 required content points; minimum 2 minutes; professional delivery.
Part B — Confidentiality	No sensitive client details, IPs, or credentials in the video.
Part C — Video 2 quality	Community-appropriate tone; covers all 5 content points; minimum 2 minutes; genuine reflection not promotional filler.
Part C — LinkedIn post	150–250 words; Video 2 linked; correct tags; no client specifics; relevant hashtags.

Submission

What You Submit

One Google Drive folder containing:

- D1 — Network Traffic Analysis Report (PDF or DOCX)
- D2 — Incident Report to CISO (PDF or DOCX)
- D3 — MITRE ATT&CK Mapping (PDF or DOCX)
- D4 Part A — Judgment Essay (PDF or DOCX)
- D4 Part B — Video 1 link (unlisted YouTube URL or Google Drive video link — must be accessible to graders)
- D4 Part C — Video 2 link (same format) + LinkedIn/X post text or screenshot

Sharing & Posting

- Folder sharing: 'Anyone with the link → Viewer'
- Video links: paste directly into the submission folder as a text file named VIDEO_LINKS.txt
- Paste folder URL on /dashboard/reports/WEEK_02 with a 75-word executive summary of your findings

Deadlines

Milestone	Date & Time
Week opens	Monday 7 July 2026
Submission deadline	Friday 11 July 2026, 18:00 WAT
Grading	Two graders read independently; super-admin tiebreaks material disagreement
Results published	Sunday 13 July 2026, 18:00 WAT

Drafts autosave every 30 seconds. Videos must be uploaded and links confirmed accessible before 18:00 WAT — broken links at deadline = not submitted.

Universal Grading Rule

Generic answers fail. Every written claim must cite a specific packet number, IP address, domain name, or external reference. Every video must demonstrate genuine engagement with this specific capture — not a rehearsed generic presentation about network security. A grader who cannot trace your analysis back to the PCAP will mark it down and move on.

— GlobalTech Cybersecurity Internship Programme Team · Week 2 · Operation Hollow Bank